

GUARDIAN

OT Network Threat Detection Platform Security Assessment Report

Report Date	June 10, 2026 13:56
	OT/ICS Security Assessment
Protocols Tested	Modbus TCP, OPC-UA, DNP3
	107
Critical Alerts	81
	v1.0

■■ CONFIDENTIAL — For Authorized Personnel Only

1. Executive Summary

The GUARDIAN OT Threat Detection Platform conducted a comprehensive security assessment of the industrial control system network. During the assessment period, GUARDIAN detected **107 security events** across multiple OT protocols including Modbus TCP and OPC-UA.

The assessment revealed significant security gaps in the OT network, including unauthorized write commands to PLC registers, rogue device detection, replay attacks, and out-of-range value injection. A total of **81 CRITICAL** severity alerts were generated, indicating immediate risk to plant operations and safety.



Key Findings:

■	CRITICAL	Modbus TCP has no authentication — any device can write to PLC registers
■	CRITICAL	Safety system registers accessible without authorization
■	HIGH	Replay attacks possible — no sequence number validation in Modbus
■	HIGH	Rogue devices can join OT network without detection
■	MEDIUM	Excessive polling detected — attacker mapping process parameters
■	LOW	OPC-UA anonymous access enabled — process data exposed

2. Alert Statistics

2.1 Alerts by Severity

Severity	Count	Percentage	Risk Level
CRITICAL	81	75.7%	Immediate Action Required
HIGH	15	14.0%	Action Required Within 24hrs
MEDIUM	11	10.3%	Action Required Within Week

2.2 Detection Rules Triggered

Rule	Count	Severity	MITRE TTP
UNAUTHORIZED_WRITE	70	CRITICAL	T0855
ROGUE_DEVICE	12	HIGH	T0846
EXCESSIVE_POLLING	11	MEDIUM	T0840
VALUE_INJECTION	11	CRITICAL	T0855
REPLAY_ATTACK	3	HIGH	T0856

2.3 Top Attacking IPs

Rank	IP Address	Alert Count	Threat Level
1	192.168.1.99	95	CRITICAL
2	192.168.136.129	7	MEDIUM
3	192.168.1.77	5	MEDIUM

3. Attack Timeline

Chronological record of all detected security events:

Timestamp	Severity	Rule	Source IP	MITRE
2026-06-08T06:15:21	HIGH	REPLAY_ATTACK	192.168.1.99	T0856 - Spoof R
2026-06-08T06:15:20	CRITICAL	VALUE_INJECTION	192.168.1.99	T0855 - Unautho
2026-06-08T06:15:19	HIGH	ROGUE_DEVICE	192.168.1.77	T0846 - Remote
2026-06-08T06:15:19	MEDIUM	EXCESSIVE_POLLING	192.168.1.99	T0840 - Network
2026-06-08T06:15:17	CRITICAL	UNAUTHORIZED_WRITE	192.168.1.99	T0855 - Unautho
2026-06-08T06:15:17	CRITICAL	UNAUTHORIZED_WRITE	192.168.1.99	T0855 - Unautho
2026-06-08T06:15:16	CRITICAL	UNAUTHORIZED_WRITE	192.168.1.99	T0855 - Unautho
2026-06-08T06:15:16	CRITICAL	UNAUTHORIZED_WRITE	192.168.1.99	T0855 - Unautho
2026-06-08T06:15:15	CRITICAL	UNAUTHORIZED_WRITE	192.168.1.99	T0855 - Unautho
2026-06-07T05:12:48	CRITICAL	VALUE_INJECTION	192.168.1.99	T0855 - Unautho
2026-06-07T05:12:47	HIGH	ROGUE_DEVICE	192.168.1.99	T0846 - Remote
2026-06-07T05:12:47	MEDIUM	EXCESSIVE_POLLING	192.168.1.99	T0840 - Network
2026-06-07T05:12:43	CRITICAL	UNAUTHORIZED_WRITE	192.168.1.99	T0855 - Unautho
2026-06-07T05:12:42	CRITICAL	UNAUTHORIZED_WRITE	192.168.1.99	T0855 - Unautho
2026-06-07T05:12:41	CRITICAL	UNAUTHORIZED_WRITE	192.168.1.99	T0855 - Unautho
2026-06-07T05:12:40	CRITICAL	UNAUTHORIZED_WRITE	192.168.1.99	T0855 - Unautho
2026-06-07T05:12:39	CRITICAL	UNAUTHORIZED_WRITE	192.168.1.99	T0855 - Unautho
2026-06-05T12:17:38	CRITICAL	VALUE_INJECTION	192.168.1.99	T0855 - Unautho
2026-06-05T12:17:38	CRITICAL	UNAUTHORIZED_WRITE	192.168.136.129	T0855 - Unautho
2026-06-05T12:17:37	HIGH	ROGUE_DEVICE	192.168.1.99	T0846 - Remote

4. OT Asset Impact Analysis

GUARDIAN's Asset Impact Analyzer maps every attack to physical OT assets and real-world consequences:

Asset	Attack Type	Safety Risk	Risk Score	Recovery Time
Valve Actuator — Feed Lin	UNAUTHORIZED_WRITE	MEDIUM	65	30min - 2hrs
Pressure Control Valve —	REPLAY_ATTACK	CRITICAL	92	4-8 hours
Motor Speed Drive — Pump	VALUE_INJECTION	HIGH	78	1-2 days
Safety Instrumented Syste	UNAUTHORIZED_WRITE	CRITICAL	99	Days/Weeks
Temperature Controller —	UNAUTHORIZED_WRITE	HIGH	85	2-4 hours

5. MITRE ATT&CK; for ICS Mapping

All detected attacks mapped to MITRE ATT&CK; for ICS framework techniques:

Detection Rule	Tactic	Technique ID	Technique Name	Severity
UNAUTHORIZED_WRITE	Impair Process Control	T0855	Unauthorized Command Message	CRITICAL
ROGUE_DEVICE	Initial Access	T0846	Remote System Discovery	HIGH
REPLAY_ATTACK	Evasion	T0856	Spoof Reporting Message	HIGH
EXCESSIVE_POLLING	Discovery	T0840	Network Connection Enumeration	MEDIUM
VALUE_INJECTION	Impair Process Control	T0855	Unauthorized Command Message	CRITICAL
HONEYPOT_TRIGGERED	Collection	T0843	Program Download	CRITICAL

6. Remediation Roadmap

6.1 Immediate Actions (0-7 days)

#	Action	Priority	Effort
1	Implement Modbus TCP authentication using vendor-specific security modules	CRITICAL	Medium
2	Enable OPC-UA security mode — disable anonymous access	CRITICAL	Low
3	Deploy network segmentation — isolate OT from IT network	CRITICAL	High
4	Implement IP whitelisting on all Modbus devices	HIGH	Low
5	Enable audit logging on all PLC and SCADA systems	HIGH	Low

6.2 Short Term (7-30 days)

#	Action	Priority	Effort
1	Deploy IDS/IPS solution for OT network monitoring	HIGH	High
2	Implement Modbus sequence number validation	HIGH	Medium
3	Deploy honeypot registers for early attacker detection	HIGH	Low
4	Conduct OT security awareness training for operators	MEDIUM	Medium
5	Implement automated backup for all PLC programs	MEDIUM	Low

6.3 Long Term (30-90 days)

#	Action	Priority	Effort
1	Implement IEC 62443 compliance framework across OT network	HIGH	Very High
2	Deploy encrypted communications for all OT protocols	HIGH	High
3	Implement Security Information and Event Management (SIEM)	MEDIUM	High
4	Conduct annual OT penetration testing exercise	MEDIUM	High
5	Develop and test OT incident response playbook	MEDIUM	Medium